

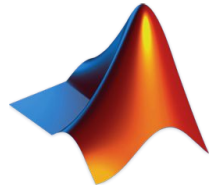
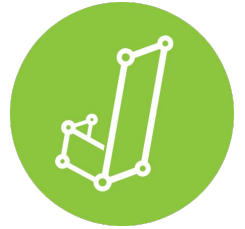
Data analysis with





Why Jamovi?

- Free alternative to SPSS
- Open-sourced, open-access
- Flexibility, similar to R (syntax mode)
- Replicability
- Beginner friendly



Overview

10:00-12:30

1. Load data
2. Data and measure types
3. Compute variables
4. Descriptive statistics
5. Data visualization
6. Filter
7. Transform variables

13:30-16:30

8. Statistics
9. Modules
10. Long format vs. wide format
11. Syntax mode



Variables

Data

Analyses

Plots

Edit



Paste



Clipboard



Edit



Setup



Compute



Transform

Variables



Weights



Add

Delete



Filters



Add

Delete

Delete

Rows

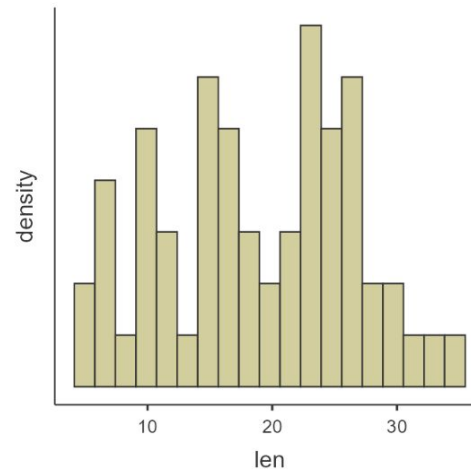
	len	supp	dose
1	4.2	VC	500
2	11.5	VC	500
3	7.3	VC	500
4	5.8	VC	500
5	6.4	VC	500
6	10.0	VC	500
7	11.2	VC	500
8	11.2	VC	500
9	5.2	VC	500
10	7.0	VC	500
11	16.5	VC	1000
12	16.5	VC	1000
13	15.2	VC	1000
14	17.3	VC	1000
15	22.5	VC	1000
16	17.3	VC	1000
17	13.6	VC	1000
18	14.5	VC	1000
19	18.8	VC	1000
20	15.5	VC	1000
21	23.6	VC	2000
22	18.5	VC	2000
23	33.9	VC	2000
24	25.5	VC	2000
25	26.4	VC	2000

Results

Descriptives

Plots

len



Zoom



100%



Results

Number format

3 sf

p-value format

3 dp

Decimal symbol

Dot

References

Visible

Syntax mode



Plots

Plot theme

Default

Color palette

1 SPSS

Import

Default missing

NA

Language

System default

Developer mode





Variables

Data

Analyses

Plots

Edit



Edit



Edit



Compute

Transform
Variables

Add



Delete

Filters
Rows

Search variables



Name

Description



len

Enter description



supp



dose

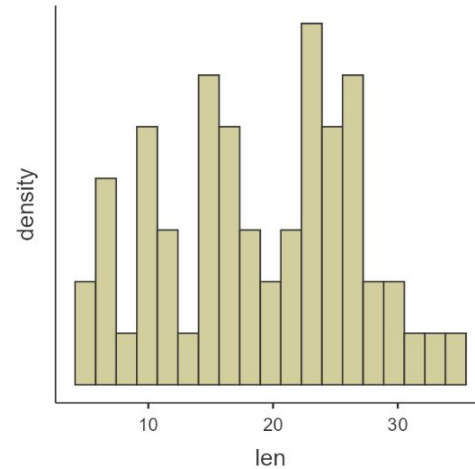


Results

Descriptives

Plots

len



1. Load data

- Open: Loading one single file
 - Data Library
- Special Import: Loading and stacking multiple files
 - for combining datasets (e.g., one dataset for each participant)



jamovi

New

Open

Special Import

Save

Save As

Export

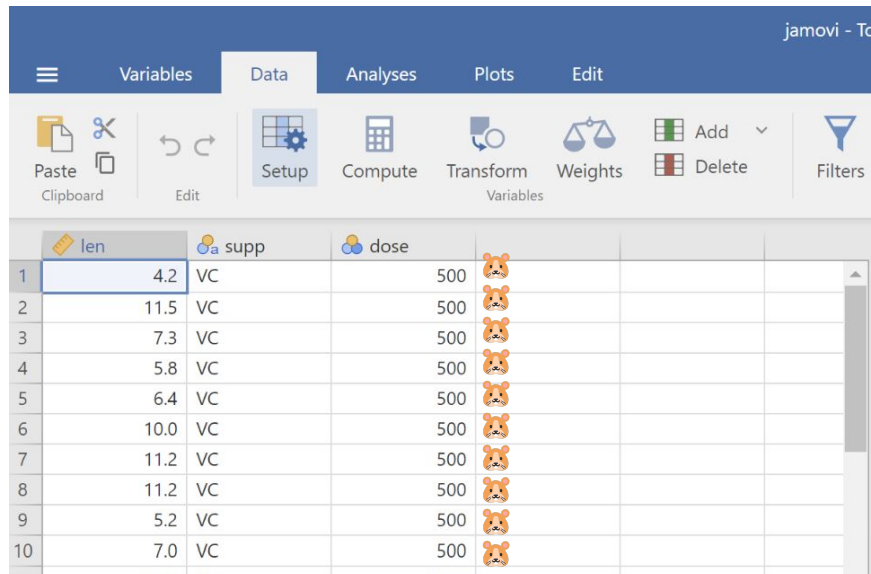


[Guinea pig - Wikipedia](#)

Exercise 1: load the Tooth Growth dataset

The **Tooth Growth** example contains the length of teeth from guinea pigs (the **len** column) fed different dosages (the **dose** column) of supplements: vitamin C or orange juice (recorded in the **supp** column as **VC** and **OJ**).

	500	1000	2000
VC 			
OJ 			



The screenshot shows the Jamovi software interface with the 'Data' tab selected. The dataset 'Tooth Growth' is loaded, displaying columns for 'len' (teeth length), 'supp' (supplement), and 'dose' (dosage). The data is organized into rows, with the first 10 rows visible. The 'len' column contains numerical values, the 'supp' column contains categorical values ('VC' and 'OJ'), and the 'dose' column contains numerical values (500, 1000, 2000). The interface also shows a toolbar with various analysis and editing tools.

	len	supp	dose			
1	4.2	VC	500			
2	11.5	VC	500			
3	7.3	VC	500			
4	5.8	VC	500			
5	6.4	VC	500			
6	10.0	VC	500			
7	11.2	VC	500			
8	11.2	VC	500			
9	5.2	VC	500			
10	7.0	VC	500			

2. Data type and measure types

Data variables can be one three data types:

Integer

Decimal

Text

and one of four measure types:



Nominal



Ordinal

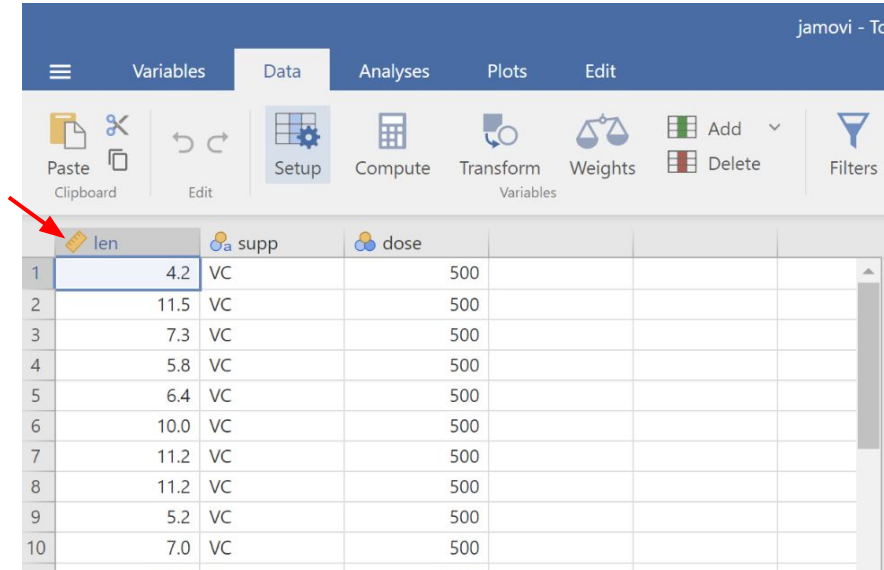


Continuous

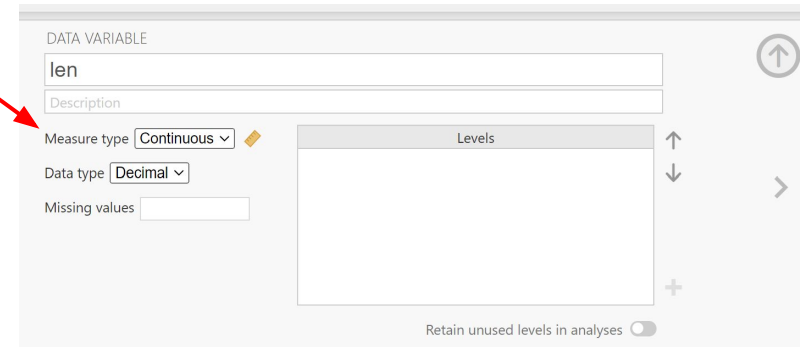


ID

They are editable (to a certain extent)



	len	supp	dose			
1	4.2	VC	500			
2	11.5	VC	500			
3	7.3	VC	500			
4	5.8	VC	500			
5	6.4	VC	500			
6	10.0	VC	500			
7	11.2	VC	500			
8	11.2	VC	500			
9	5.2	VC	500			
10	7.0	VC	500			



DATA VARIABLE

len

Description

Measure type **Continuous**

Data type **Decimal**

Missing values

Levels

Retain unused levels in analyses

3. Compute variables

E.g.,

- Arithmetic:
 - $A + B$
 - CRT-SRT
- Functions (across columns):
 - $\text{MEAN}(A, B)$
 - $\text{LOG}_{10}(\text{len})$
- Variable function (across rows):
 - $\text{VMEAN}(\text{len}, \text{group_by}=\text{dose})$
- Statistics, e.g., z-scores
 - $(\text{dose} - \text{VMEAN}(\text{dose})) / \text{VSTDEV}(\text{dose})$
 - $Z(\text{dose})$

Exercise 2: Label and compute variables

1. Rename *len* to *tooth_length*
2. Label *OJ* as *orange_juice*, and *VC* as *vitamin_c*
3. Change the variable type of *dose* into ordinal
4. Reorder the levels of *dose* as 2000, 1000, 500 (instead of 500, 1000, 2000)
5. Label the levels of *dose* in mg (Hint: 1mg = 1000 mcg)
6. Compute *len_z* the z-score of *len* (Hint: $Z(\text{tooth_length})$)

4. Descriptive statistics

Descriptives

Variables

tooth_length

Split by

supp
dose

Descriptives Variables across rows ☐ Frequency tables

Statistics

Sample Size

☒ N ☒ Missing

Percentile Values

☐ Cut points for 4 equal groups☐ Percentiles 25,50,75

Dispersion

☒ Std. deviation ☒ Minimum☐ Variance ☒ Maximum☐ Range ☐ IQR

Mean Dispersion

☐ Std. error of Mean☐ Confidence interval for Mean 95 %

Central Tendency

☒ Mean☒ Median☐ Mode☐ Sum

Distribution

☐ Skewness☐ Kurtosis

Normality

☐ Shapiro-Wilk

Outliers

☐ Most extreme 5 values

Plots

Histograms

☐ Histogram☐ Density

Q-Q Plots

Box Plots

☐ Box plot☒ Label outliers☐ Violin

Bar Plots

☐ Bar plot

Descriptives

Descriptives

	supp	dose	N	Missing	Mean	Median	SD	Minimum	Maximum
tooth_length	orange_juice	0.5	10	0	13.23	12.25	4.46	8.20	21.5
		1	10	0	22.70	23.45	3.91	14.50	27.3
		2	10	0	26.06	25.95	2.66	22.40	30.9
vitamin_c		0.5	10	0	7.98	7.15	2.75	4.20	11.5
		1	10	0	16.77	16.50	2.52	13.60	22.5
		2	10	0	26.14	25.95	4.80	18.50	33.9

References

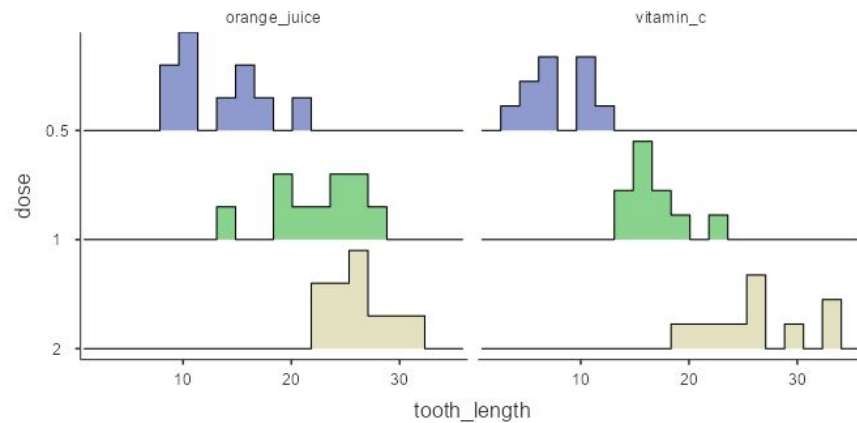
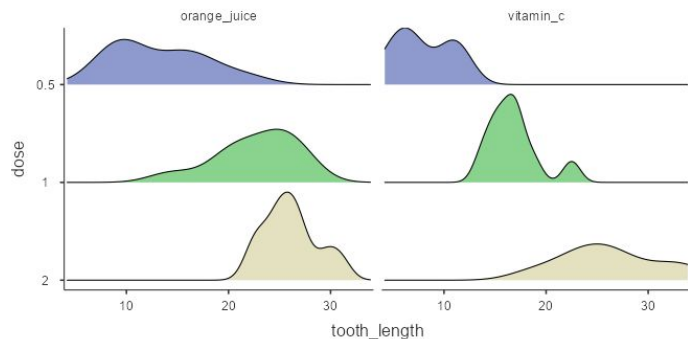
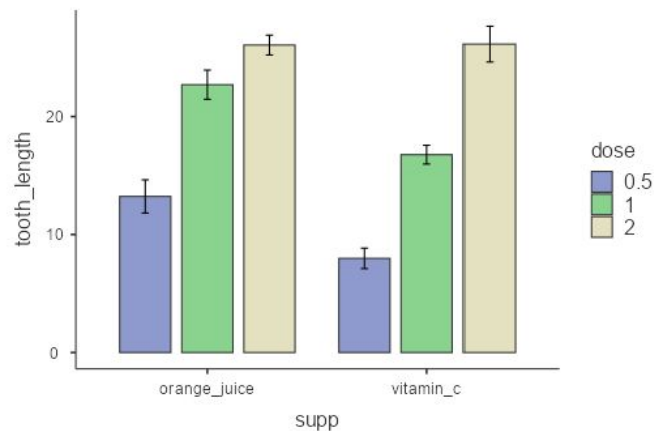
[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from <https://www.jamovi.org>.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2025-05-25).

5. Data visualization

- bar charts
- histograms
- density plots

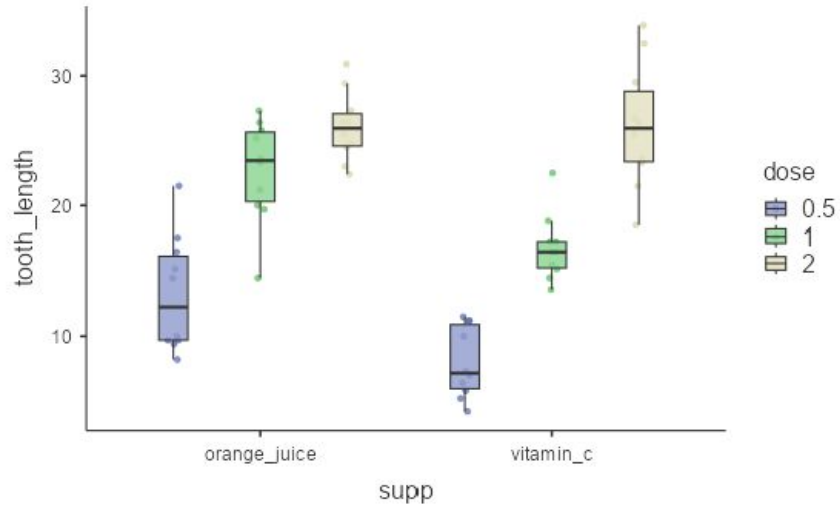
tooth_length



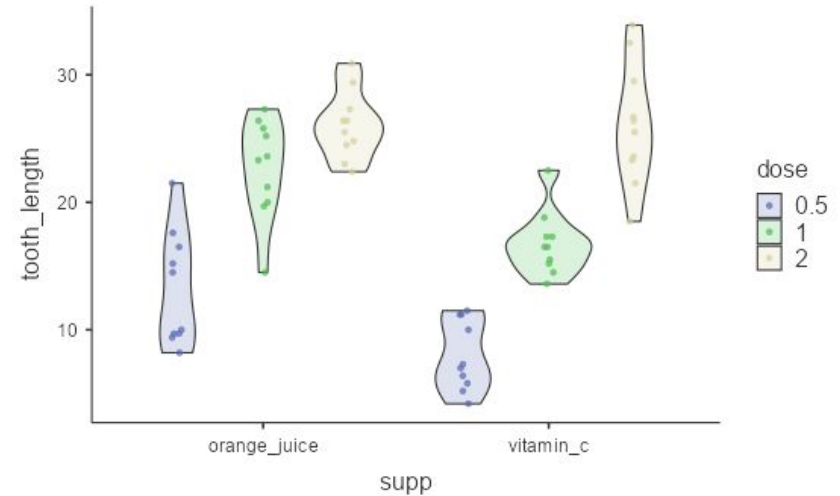
5. Data visualization

- box plots
- violin plots

tooth_length



tooth_length

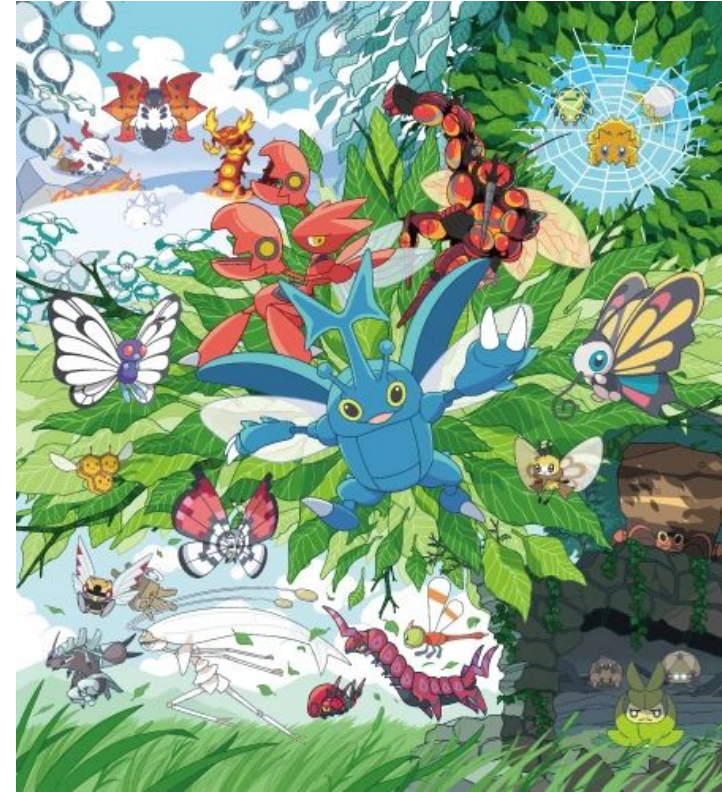


Exercise 3: Bugs!

The extent to which men and women want to kill arthropods that vary in frighteningness (low, high) and disgustingness (low, high). Each participant rates their attitudes towards all arthropods.

With 372 rows and 6 variables

- subject. Subject ID
- gender. Female or Male
- region. Region of the world the participant was from.
- education. Level of education.
- desire. The desire to kill an arthropod indicated on a scale from 0 to 10, In the following conditions:
 - **LDLF**: low frighteningness and low disgustingness;
 - **LFHD**: low frighteningness and high disgustingness;
 - **HFHD**: high frighteningness and low disgustingness;
 - **HFHD**: high frighteningness and high disgustingness



[Bug \(Pokémon type\) | Nintendo | Fandom](#)

https://www.indrapatil.com/ggstatsplot/reference/bugs_long.html

Exercise 3: Bugs!

1. Load the Bugs dataset from the data library
2. Check the data and measure types; correct them if needed
3. Compute a *desire_to_kill* rating by averaging the ratings across all conditions
4. Explore the summary statistics of this *desire_to_kill* variable
5. Based on data visualization, determine if *desire_to_kill* differs across:
 - a. Gender
 - b. Education
 - c. Region
6. Categorize individuals (rows) into high vs. low *desire_to_kill*

6. Filter

- With a clear rationale, outliers / invalid data could be excluded from the analysis
- E.g., in the Bugs dataset, subject 11 has very low ratings across all conditions
 - 0, 2.5, 0, 0
- It could be excluded with the filter function
- As with other changes in the data, outputs will be automatically updated



Variables

Data

Analyses

Plots

Edit

Paste
Clipboard

6. Filter



Edit



Compute

Transform
Variables

Weights



Add



Delete



Filters



Add



Delete

Rows

ROW FILTERS



Filter 1

active ☒ f_x

= Subject != 11

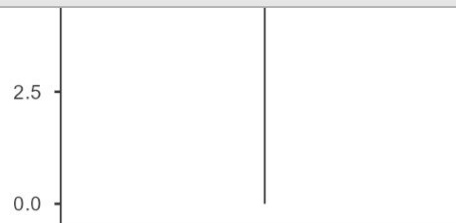
!= means "not equal"



Description



	Filter 1	Subject	Gender	Region	Education	LDLF
2	✓	2	Female	North America	advance	1
3	✓	3	Female	Europe	college	
4	✓	4	Female	North America	college	
5	✓	5	Female	North America	some	
6	✓	6	Female	Europe	some	
7	✓	7	Female	North America	some	1
8	✓	8	Female	North America	high	1
9	✓	9	Female	North America	high	
10	✓	10	Female	Other	high	
11	✗	11	Male	North America	some	



HDHF

7. Transform variables

- Unlike *compute*, which could work on multiple variables, *transform* works on one variable. However, it could reuse transformation functions.
- Typically used for recoding variables, or with functions applied to multiple variables repeatedly
- E.g., median split could be applied to dichotomize variables (usually not recommended though)
 - In the BUGS dataset, an individual could be categorized as “*killer*” if the *desire_to_kill* rating is larger or equal to the median, and “*not killer*” if not



Variables

Data

Analyses

Plots

Edit

Paste
Clipboard

Edit



Transform



Compute



Variables



Add



Delete



Filters



Add



Delete

Rows

7. Transform variables

TRANSFORMED VARIABLE

killer

desire_to_kill greater than or equal to median

Source variable

desire_to_kill

using transform

median_split

Edit...

Retain unused levels in analyses ☐

	HDLF	HDHF	desire_to_...	killer
20	8.5	9.5	8.500	killer
21	5.0	9.5	5.500	not killer
22	4.5	7.5	6.000	not killer
23	9.0	10.0	9.750	killer
24	4.0	1.5	1.750	not killer
25	4.5	5.5	2.750	not killer
26	3.5	10.0	6.500	not killer
27	10.0	10.0	7.875	killer
28	8.0	8.0	8.250	killer
29	7.5	6.0	6.625	not killer
30	6.5	10.0	7.750	killer



7. Transform variables

TRANSFORMED VARIABLE

TRANSFORM ● used by 1

median_split

Description

Variable suffix



Measure type Auto ▼

8. Statistics!

- Unlike SPSS, Javmovi does not have as many analysis options
- However, the outputs of the analysis are often clearer and follow the APA format
- By design, the ways to execute them are similar to SPSS

8. Statistics! - Independent t test

Does *desire_to_kill* differ across genders?

Independent Samples T-Test

Independent Samples T-Test

						95% Confidence Interval		95% Confidence Interval				
		Statistic	df	p	Mean difference	SE difference	Lower	Upper		Effect Size	Lower	Upper
desire_to_kill	Student's t	0.998	85.0	.321	0.487	0.488	-0.483	1.46	Cohen's d	0.227	-0.221	0.673

Note. $H_a: \mu_{\text{Female}} \neq \mu_{\text{Male}}$

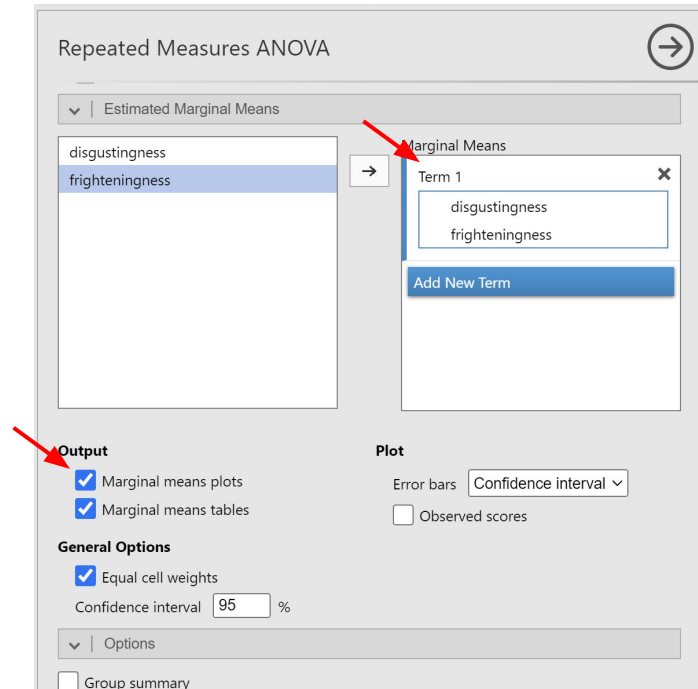
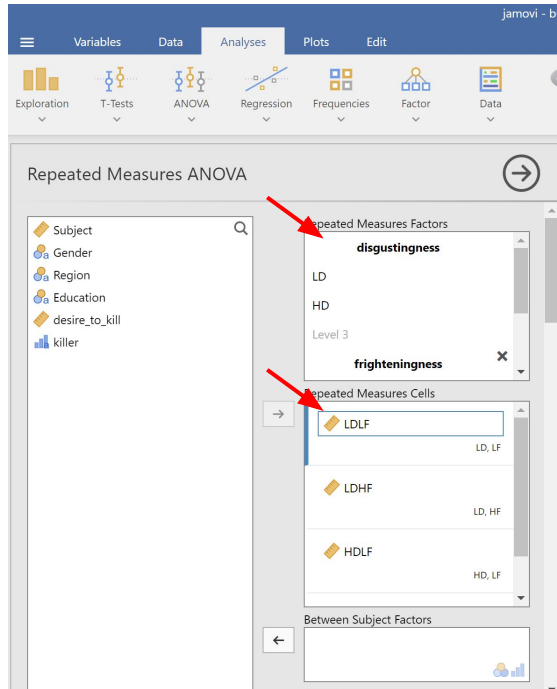
Group Descriptives

	Group	N	Mean	Median	SD	SE
desire_to_kill	Female	58	7.02	7.56	2.24	0.294
	Male	29	6.53	6.88	1.95	0.362

The screenshot shows the Jamovi software interface for an Independent Samples T-Test. The 'Analyses' tab is active. The 'Independent Samples T-Test' dialog box is open. The 'Dependent Variables' box contains 'desire_to_kill'. The 'Grouping Variable' box contains 'Gender'. Under 'Tests', 'Student's' is selected. Under 'Hypothesis', 'Group 1 ≠ Group 2' is selected. Under 'Missing values', 'Exclude cases analysis by analysis' is selected. Under 'Additional Statistics', 'Mean difference', 'Confidence interval 95%', 'Effect size', 'Confidence interval 95%', and 'Descriptives' are all checked. Red arrows point to the 'Dependent Variables', 'Grouping Variable', and 'Missing values' sections.

8. Statistics! - Repeated measures ANOVA

Do *disgustingness* and *frighteningness* affect *desire_to_kill*?



8. Statistics! - Repeated measures ANOVA

Repeated Measures ANOVA

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
disgustingness	48.75	1	48.75	12.18	< .001
Residual	348.37	87	4.00		
frighteningness	177.56	1	177.56	41.63	< .001
Residual	371.07	87	4.27		
disgustingness * frighteningness	6.55	1	6.55	2.15	.146
Residual	264.58	87	3.04		

Note. Type 3 Sums of Squares

[3]

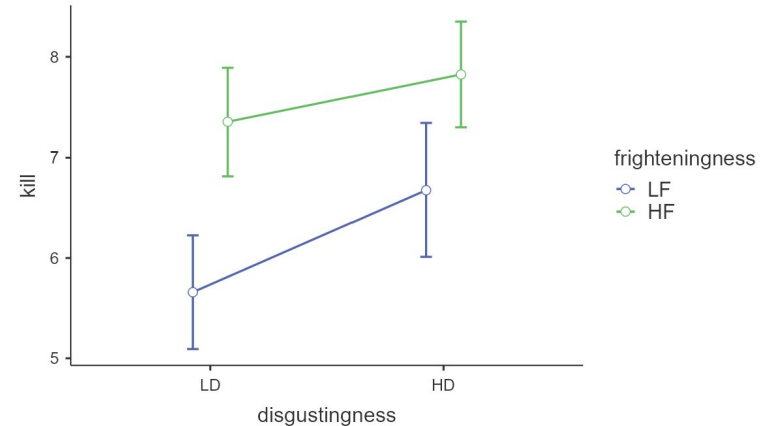
Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Residual	1595	87	18.3		

Note. Type 3 Sums of Squares

Estimated Marginal Means

disgustingness * frighteningness



8. Statistics! - Correlation

Are the ratings across four conditions correlated?

jamovi - bu

Variables Data Analyses Plots Edit

Exploration T-Tests ANOVA Regression Frequencies Factor Data

Correlation Matrix

Subject
desire_to_kill
killer
Gender
Region
Education

←

- HDHF
- HDLF
- LDHF
- LDLF

Correlation Coefficients

- ☒ Pearson
- ☐ Spearman
- ☐ Kendall's tau-b

Additional Options

- ☒ Report significance
- ☐ Flag significant correlations
- ☐ N
- ☐ Confidence intervals
Interval 95 %

Hypothesis

- ☒ Correlated
- ☐ Correlated positively
- ☐ Correlated negatively

Plot

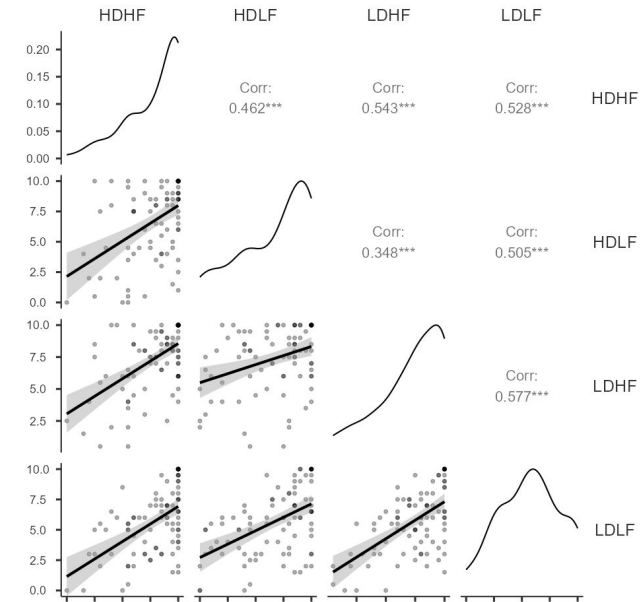
- ☒ Correlation matrix
- ☒ Densities for variables
- ☒ Statistics

Correlation Matrix

Correlation Matrix

		HDHF	HDLF	LDHF	LDLF
HDHF	Pearson's r	—			
	df	—			
	p-value	—			
HDLF	Pearson's r	0.462	—		
	df	88	—		
	p-value	< .001	—		
LDHF	Pearson's r	0.543	0.348	—	
	df	88	87	—	
	p-value	< .001	< .001	—	
LDLF	Pearson's r	0.528	0.505	0.577	—
	df	90	89	89	—
	p-value	< .001	< .001	< .001	—

Plot



8. Statistics! - Regression

Do gender, region, and education predict overall *desire_to_kill*?

Linear Regression

Subject

LDLF

LDHF

HDLF

HDHF

killer

Dependent Variable

→ desire_to_kill

Covariates

Factors

← Gender

Region

Education

Weights (optional)

→

Linear Regression

> | Model Fit

> | Model Coefficients

▼ | Estimated Marginal Means

Gender

Region

Education

→ Marginal Means

Term 1

Gender

Region

Education

Add New Term

General Options

☒ Equal cell weights

☒ Confidence interval

Interval 95 %

Output

☒ Marginal means plots

☐ Marginal means tables

Linear Regression

Model Fit Measures

Model	R	R ²
1	0.399	0.159

Note. Models estimated using sample size of N=86

Model Coefficients - desire_to_kill

Predictor	Estimate	SE	t	p
Intercept*	3.093	2.982	1.037	.303
Gender:				
Male – Female	–0.500	0.490	–1.020	.311
Region:				
Europe – Australia	–0.967	2.245	–0.431	.668
North America – Australia	0.782	2.125	0.368	.714
Other – Australia	1.833	2.415	0.759	.450
South America – Australia	3.875	2.958	1.310	.194
Education:				
college – advance	3.800	2.322	1.636	.106
high – advance	4.000	2.183	1.832	.071
less – advance	3.518	2.149	1.637	.106
partial – advance	2.250	2.573	0.874	.385
some – advance	2.907	2.125	1.368	.175

* Represents reference level

9. Modules

Frequencies

Factor

Base R

PPDA

medmod

Survival

Walrus

Statkat

R

Linear Models

MAJOR

distrACTION

Flexplot

Modules

jamovi



jamovi library

Manage installed

Installed Modules

Installed

Available

Sideload



Rj - Editor to run R code inside jamovi 1.0.8

Jonathon Love

Provides an editor allowing you to enter R code, and analyse your data using R inside jamovi. Note that when using the 'System R', Rj is currently not compatible with R 3.5 or newer.

INSTALL



jpower - Power Analysis for Common Research Designs 0.1.2

Richard D. Morey, Ravi Selker

Power analysis for common research designs.

INSTALL



GAMLj - General Analyses for Linear Models 2.0.0

Marcello Gallucci

A suite for estimation of linear models, such as the general linear model, linear mixed model, generalized linear models and generalized mixed models. For each family, models can be estimated with categorical and/or continuous variables, with options to facilitate estimation of interactions, simple slopes, simple effects, post-hoc tests, contrast analysis and visualization of the results.

gamlj.github.io

INSTALLED

jmv
jamoviStatkat
Method Selection TooljmvbaseR
Base Rjamm
Advanced Mediation Modelsgamlj
General Analyses for Linear Models in Jamoviflexplot
Graphically Based Data AnalysisprocessR
R-Adaptation of Andrew Hayes' PROCESS-Ma...distrACTION
Quantiles and Probabilities of Continuous an...Rj
Editor to run R code inside jamovipsychoPDA
Psychometrics & Post-Data Analysis

medmod

- Rj
- jTransform

10. Long format and wide format

- Data are often stored in the long or wide format
- E.g.,
 - behavioral tasks with multiple trials store data in the long format
 - statistical packages sometimes expect data in the wide format
- Jamovi doesn't have a function for that but the module *jTransfrom* can do that

The screenshot shows the 'Long to Wide' transformation module in Jamovi. On the left, a list of variables includes 'position', 'trial_resp.', and 'trial_rt', with 'trial_rt' selected. On the right, four categories of variables are shown: 'Variables That Identify the Same Unit' (containing 'ID'), 'Variables That Differentiate Within a Unit' (containing 'direction' and 'position_text'), 'Variables To Be Transformed' (containing 'trial_accuracy'), and 'Variables NOT To Be Transformed' (empty). At the bottom, there are settings for 'Organization of Variables in the Output' (set to 'Time Steps Are Adjacent'), 'How to Accumulate?' (set to 'Calculate Mean'), and a 'Separator' (set to '_'). A 'Create' button is at the bottom right, and a 'Show Help / Information' checkbox is at the bottom left.

Long to Wide

position
trial_resp.
trial_rt

Variables That Identify the Same Unit
ID

Variables That Differentiate Within a Unit
direction
position_text

Variables To Be Transformed
trial_accuracy

Variables NOT To Be Transformed

Organization of Variables in the Output: Time Steps Are Adjacent
How to Accumulate?: Calculate Mean
Separator: _

☒ Show Help / Information

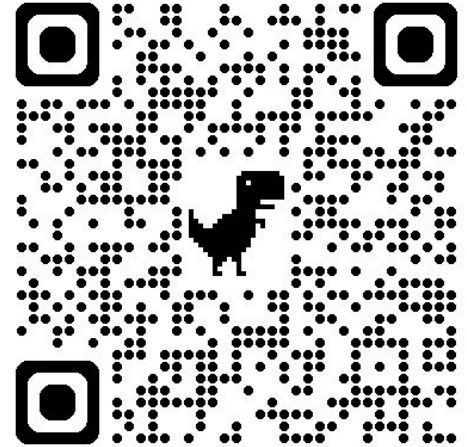
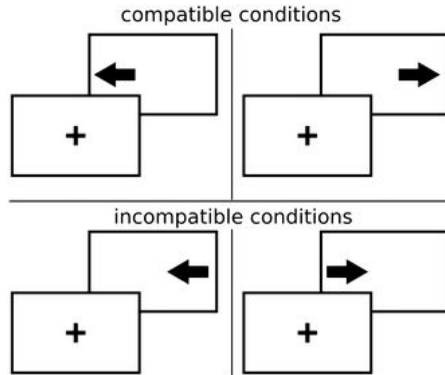
Create

Exercise 4: Simon task

Download the data sets *simon_data* and *gonogo_data* from this link

https://drive.google.com/drive/folders/12meW_MLh03t6F_aCJFA02S9Xd0tXZd_B?usp=drive_link

Open *simon_data* and *Special Import* all files



Exercise 4: Simon task

1. Convert positions coordinates to “left” or “right” (HINT: $[-0.25, 0]$ = “left”)
2. Compute *trial_accuracy* (HINT: `VALUE((direction == trial_resp.))`)
3. Convert the long format data file into wide format, for accuracy and conditions respectively, and merge them into one

Accuracy by condition (direction × position)

ID	direction	position	trial_resp.	trial_rt	trial_acc...	position_...
1	1 right	[0.25, 0]	right	0.369	1	right
2	1 right	[0.25, 0]	right	0.391	1	right
3	1 left	[0.25, 0]	left	0.468	1	right
4	1 right	[0.25, 0]	left	0.393	0	right
5	1 right	[0.25, 0]	right	0.351	1	right
6	1 left	[-0.25, 0]	right	0.390	0	left
7	1 left	[0.25, 0]	right	0.456	0	right
8	1 right	[0.25, 0]	right	0.497	1	right
9	1 right	[-0.25, 0]	right	0.486	1	left
10	1 right	[-0.25, 0]	right	0.504	1	left
11	1 right	[0.25, 0]	right	0.449	1	right
12	1 left	[0.25, 0]	left	0.410	1	right

Long format:
Each measure is a row

ID	trial_acc...	trial_acc...	trial_acc...	trial_acc...
1	0.92	0.88	0.92	0.96
2	0.84	0.88	0.84	0.92
3	0.84	0.92	0.88	0.88
4	0.88	0.96	0.92	0.92
5	0.96	0.92	0.92	0.92
6	0.92	0.96	0.84	0.88
7	0.96	0.88	0.84	0.88
8	0.92	0.92	0.84	0.92
9	0.88	0.88	0.96	0.92
10	0.88	0.92	0.92	0.84

Wide format:
Each participant is a row

RT by condition (direction × position × trial_accuracy)

ID	direction	position	trial_resp.	trial_rt	trial_acc...	position_...
1	1	right	[0.25, 0]	right	0.369	1 right
2	1	right	[0.25, 0]	right	0.391	1 right
3	1	left	[0.25, 0]	left	0.468	1 right
4	1	right	[0.25, 0]	left	0.393	0 right
5	1	right	[0.25, 0]	right	0.351	1 right
6	1	left	[-0.25, 0]	right	0.390	0 left
7	1	left	[0.25, 0]	right	0.456	0 right
8	1	right	[0.25, 0]	right	0.497	1 right
9	1	right	[-0.25, 0]	right	0.486	1 left
10	1	right	[-0.25, 0]	right	0.504	1 left
11	1	right	[0.25, 0]	right	0.449	1 right
12	1	left	[0.25, 0]	left	0.410	1 right

Long format:
Each measure is a row

Wide format:
Each participant is a row

ID	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...
1	1	0.513	0.504	0.390	0.405	0.440	0.442	0.495	0.476
2	2	0.555	0.520	0.314	0.345	0.373	0.376	0.411	0.430
3	3	0.463	0.455	0.442	0.384	0.436	0.386	0.471	0.478
4	4	0.463	0.497	0.384	0.438	0.416	0.426	0.541	0.483
5	5	0.392	0.461	0.401	0.398	0.365	0.433	0.517	0.512
6	6	0.476	0.507	0.464	0.461	0.377	0.460	0.438	0.501
7	7	0.422	0.473	0.347	0.381	0.463	0.417	0.464	0.503
8	8	0.385	0.497	0.345	0.440	0.437	0.424	0.441	0.487
9	9	0.521	0.446	0.395	0.369	0.295	0.388	0.397	0.429
10	10	0.507	0.507	0.396	0.406	0.438	0.393	0.479	0.508

Merge

Merge (Add Columns) →

trial_accuracy_left_right

trial_accuracy_left_left

trial_accuracy_right_right

trial_accuracy_right_left

→

Variable(s) to Match the Data Sets By

ID

Data Set(s) to Add: /Users/yihongko/Downloads/jamovi_workshop/simc Browse...

Type of Merging Operation: Keeps All Cases (Rows) ▼

☐ Show Help / Information Create

	ID	trial_acc...	trial_acc...	trial_acc...	trial_acc...	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...
1	1	0.92	0.88	0.92	0.96	0.513	0.504	0.390	0.405	0.440	0.442	0.495	0.476
2	10	0.88	0.92	0.92	0.84	0.507	0.507	0.396	0.406	0.438	0.393	0.479	0.508
3	2	0.84	0.88	0.84	0.92	0.555	0.520	0.314	0.345	0.373	0.376	0.411	0.430
4	3	0.84	0.92	0.88	0.88	0.463	0.455	0.442	0.384	0.436	0.386	0.471	0.478
5	4	0.88	0.96	0.92	0.92	0.463	0.497	0.384	0.438	0.416	0.426	0.541	0.483
6	5	0.96	0.92	0.92	0.92	0.392	0.461	0.401	0.398	0.365	0.433	0.517	0.512
7	6	0.92	0.96	0.84	0.88	0.476	0.507	0.464	0.461	0.377	0.460	0.438	0.501
8	7	0.96	0.88	0.84	0.88	0.422	0.473	0.347	0.381	0.463	0.417	0.464	0.503
9	8	0.92	0.92	0.84	0.92	0.385	0.497	0.345	0.440	0.437	0.424	0.441	0.487
10	9	0.88	0.88	0.96	0.92	0.521	0.446	0.395	0.369	0.295	0.388	0.397	0.429

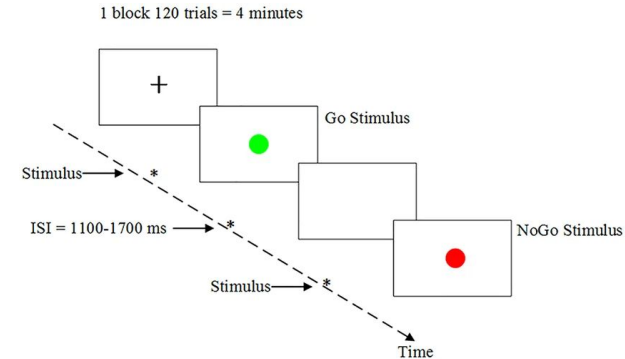
Exercise 4: Does congruence affect accuracy and RT?

1. Compute a variable that indicates congruence
2. Create a wide format of the data
3. Use a statistical test to determine if there's any difference between congruence conditions in accuracy or RT

Final challenge!

Ten students completed a go/no-go task. To analyze their data, you need to:

1. Preprocess data
 - Import data
 - Rename variables and values if necessary
 - Compute trial_accuracy
 - Compute trial_outcome (Hit, FA, Miss, CR)
2. Data analysis
 - Does accuracy differ across go and no-go condition?
 - Does RT differ across hits and false alarm?
 - Find out how to calculate d' and criterion c
 - Are the values of d' and c significantly different from 0?



Guo, Z., Chen, R., Liu, X., Zhao, G., Zheng, Y., Gong, M., & Zhang, J. (2018). The impairing effects of mental fatigue on response inhibition: An ERP study. *PLoS one*, 13(6), e0198206.

How to compute *trial_outcome*

COMPUTED VARIABLE

trial_outcome

Description

Formula

f_x

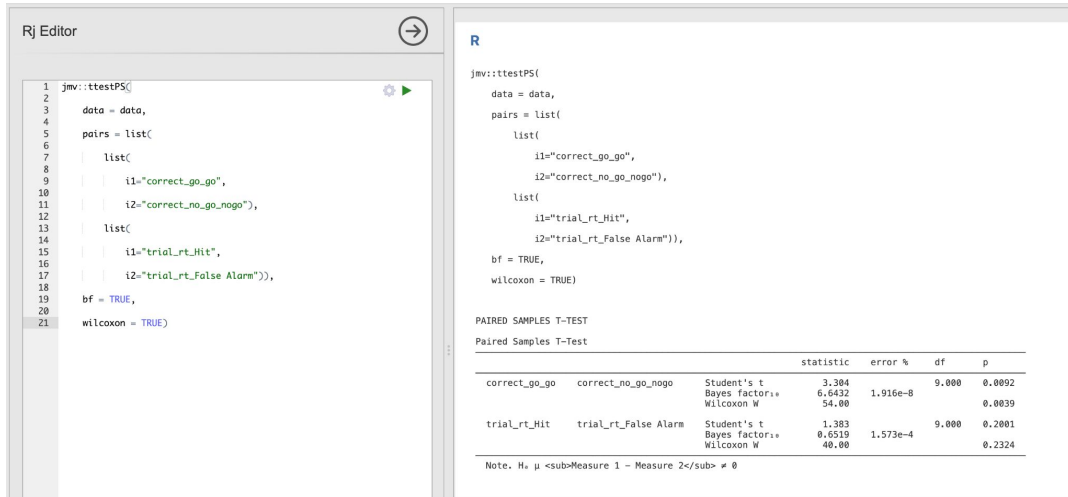
```
= IF(trial_type == 'go' and trial_resp. == 'press',  
'Hit', IF(trial_type == 'go' and trial_resp. == 'none',  
'Miss', IF(trial_type == 'nogo' and trial_resp. ==  
'none', 'Correct Rejection', 'False Alarm')))
```

Retain unused levels in analyses ☐

11. Jamovi syntax in R

Procedures on Jamovi could be recreated using the underlying R codes, which is available if Syntax mode enabled.

Find some code above the outputs in the result panel, and copy that to the Rj editor and run it!



Rj Editor

```
1 jmv::ttestPS(  
2   data = data,  
3   pairs = list(  
4     list(  
5       i1="correct_go_go",  
6       i2="correct_no_go_nogo"),  
7     list(  
8       i1="trial_rt_Hit",  
9       i2="trial_rt_False Alarm")  
10    ),  
11    bf = TRUE,  
12    wilcoxon = TRUE)  
13 )
```

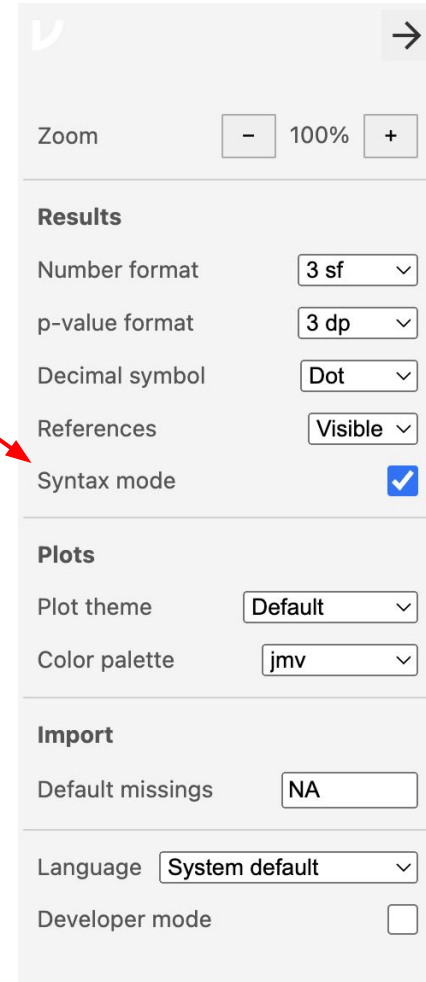
R

```
jmv::ttestPS(  
  data = data,  
  pairs = list(  
    list(  
      i1="correct_go_go",  
      i2="correct_no_go_nogo"),  
    list(  
      i1="trial_rt_Hit",  
      i2="trial_rt_False Alarm")  
    ),  
    bf = TRUE,  
    wilcoxon = TRUE)  
)
```

PAIRED SAMPLES T-TEST

			statistic	error %	df	p
correct_go_go	correct_no_go_nogo	Student's t	3.384		9.000	0.0092
		Bayes factor	6.6432	1.916e-8		
		Wilcoxon W	54.00			0.0039
trial_rt_Hit	trial_rt_False Alarm	Student's t	1.303		9.000	0.2001
		Bayes factor	0.6519	1.573e-4		
		Wilcoxon W	40.00			0.2324

Note. N = μ _{Measure 1 - Measure 2} ≠ 0



Results

Number format: 3 sf

p-value format: 3 dp

Decimal symbol: Dot

References: Visible

Syntax mode: ☒

Plots

Plot theme: Default

Color palette: jmv

Import

Default missingings: NA

Language: System default

Developer mode: ☐

If you're interested in R

R: [CRAN](#)



RStudio: [RStudio IDE | The Premier Free IDE for R](#)



RStudio is an Integrated Development Environment (IDE) for R

Resources

<https://youtu.be/mZomeS0tLxY?si=YBllvCYcFIOqeWh3>

<https://lsj.readthedocs.io/en/latest/>

[community resources - jamovi](#)

[Statistics in jamovi – Simple Book Publishing](#)

[jamovi Documentation](#)

